

BIOMINER: A Multi-modal System for Automated Mining of Protein-Ligand Bioactivity Data from Literature

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ABSTRACT

Protein-ligand bioactivity data published in literature are essential for drug discovery, yet manual curation struggles to keep pace with rapidly growing literature. Automated bioactivity extraction is challenging due to the multi-modal distribution of information (text, tables, figures, structures) and the complexity of chemical representations (e.g., Markush structures). Furthermore, the lack of standardized benchmarks impedes the evaluation and development of extraction methods. In this work, we introduce BIOMINER, a multi-modal system designed to automatically extract protein-ligand bioactivity data from thousands to potentially millions of publications. BIOMINER employs a modular, agent-based architecture, leveraging a synergistic combination of multi-modal large language models, domain-specific models, and domain tools to navigate this complex extraction task. To address the benchmark gap and support method development, we establish BIOVISTA, a comprehensive benchmark comprising 16,457 bioactivity entries and 8,735 chemical structures curated from 500 publications. On BIOVISTA, BIOMINER validates its extraction ability and provides a quantitative baseline, achieving F1 scores of 0.22, 0.45, and 0.53 for bioactivity triplets, chemical structures, and bioactivity measurement with high throughput (14s/paper on 8 V100 GPUs). We further demonstrate BIOMINER's practical utility via three applications: (1) extracting 67,953 data from 11,683 papers to build a pre-training database that improves downstream models performance by 3.1%; (2) enabling a human-in-the-loop workflow that doubles the number of high-quality NLRP3 bioactivity data, helping 38.6% improvement over 28 QSAR models and identification of 16 hit candidates with novel scaffolds; and (3) accelerating the annotation of the protein-ligand structures in PoseBusters benchmark with reported bioactivity, achieving a 5-fold speed increase and 10% accuracy improvement over manual methods. BIOMINER and BIOVISTA provide a scalable extraction methodology and a rigorous benchmark, paving the way to unlock vast amounts of previously inaccessible bioactivity data and accelerate data-driven drug discovery. All codes and data are available at [GitHub](#).

Main

Protein-ligand bioactivity data represents a cornerstone of modern drug discovery¹. Quantifying the interaction strength between potential drug molecules (ligands) and their biological targets (proteins), this information provides fundamental insights into structure-activity relationships (SAR) and the core principles of medicinal chemistry²⁻⁷. Crucially, it serves as the essential fuel for developing and validating predictive computational models – from traditional quantitative structure-activity relationship (QSAR) models to the artificial intelligence (AI) algorithms increasingly employed for virtual screening and hit identification^{8,9}. Access to comprehensive and accurate bioactivity data is therefore indispensable for accelerating the discovery and design of novel therapeutics. Over the past several years, there has been a considerable increase in experimentally determined bioactivity data, meticulously curated and made publicly accessible through databases such as ChEMBL¹⁰, BindingDB¹¹, PDBbind¹², and PubChem¹³. These invaluable resources serve as foundational pillars for drug discovery efforts.

However, the construction of these datasets relies on a labor-intensive process, requiring domain experts to painstakingly identify and transcribe bioactivity data from a vast number of scientific publications. This manual curation process can often take hours per publication, representing a significant bottleneck in scaling up bioactivity databases. With the exponential growth of scientific literature, a substantial portion of valuable bioactivity data remains inaccessible and unstructured, highlighting the urgent need for robust and efficient automated tools capable of extracting bioactivity data at scale.

Automated data extraction tools have been developed in various scientific domains, yet a critical gap exists in specialized tools tailored for the extraction of protein-ligand bioactivity data. Early tools, constrained by the limitations of prior natural language processing (NLP) and computer vision (CV) techniques, primarily focused on tasks of extracting fundamental chemical information, such as named entity recognition (NER)^{4,14} and optical chemical structure recognition (OCSR)¹⁵. Recently, the development of large language models (LLMs) advances some more promising text-mining tools. For example, GPT-based tools have been proposed for extracting metal-organic framework (MOF) synthesis conditions and enzyme-substrate interactions from literature^{16,17}. However, these methods, while demonstrating progresses, are not yet equipped to handle the extraction of protein-ligand bioactivity data.

Specifically, given a literature, the goal of bioactivity extraction is to extract all protein-ligand-bioactivity triplets (protein name, ligand SMILES, bioactivity value). Three unique challenges distinct from prior extraction tasks are presented. **Firstly**, the task is fundamentally multi-modal. Bioactivity data are distributed across different modalities, including text, tables, figures, and crucial chemical structures. This multi-modal challenge requires a sophisticated approach capable of cross-modal reasoning and information fusion, going beyond simple text mining or image mining. **Secondly**, the accurate recognition and conversion of chemical structures, particularly the widely used Markush structures (which represent groups of related chemical compounds)¹⁸, remains a significant obstacle. While existing OCSR methods have ability to recognize some Markush scaffold or R-group substitute structures, the crucial step of enumerating the specific, individual compounds they represent – a necessity for precise bioactivity data extraction – has been largely unexplored. **Finally**, besides these intrinsic data complexities, the field lacks standardized, large-scale benchmarks. This critical gap severely hinders the rigorous evaluation, comparative analysis, and systematic advancement of automated methodologies, impeding the development and validation of robust, generalizable solutions capable of reliably unlocking the wealth of bioactivity data embedded within the scientific literature.

To overcome these challenges, we propose **BIOMINER**, a multi-modal intelligent system specialized for automatically extracting protein-ligand bioactivity data from literature, and establish a comprehensive and challenging benchmark **BIOVISTA**. We first describe BIOMINER, which employs a modular, agent-based architecture to navigate this complex extraction task (Figure 1(a)). Rather than training a single end-to-end extraction model, BIOMINER decomposes the extraction process into manageable sub-tasks: document parsing (data preprocessing), chemical structure extraction, bioactivity measurement extraction, and cross-modal data integration (data postprocessing). A synergistic combination of general multi-modal large language models (MLLMs), domain-specific models (DSMs), and domain tools (DTs) is employed in the framework. Particularly, for chemical structure extraction, the system incorporates a MLLM-DSM-DT collaborative strategy specifically designed to resolve and enumerate complex Markush structures into specific full molecular structures—a pivotal step previously unaddressed at scale for automated bioactivity extraction (Figure 1(b)).

To provide standardized resources for evaluation and support future research, we introduce benchmark **BIOVISTA**, to our knowledge, the largest benchmark dedicated to protein-ligand bioactivity extraction. Meticulously curated by domain experts, BIOVISTA comprises 16,457 bioactivity entries and 8,735 unique chemical structures extracted from 500 recent publications indexed in PDBbind v2020¹². The data originates from diverse modalities within the source papers: text (15.8%), figures (11.6%), and tables (72.5%) (details in Figure S1), with a substantial fraction of chemical structures (48.7%) derived from challenging Markush representations. BIOVISTA supports six distinct evaluation tasks, ranging from end-to-end bioactivity extraction to component-level Markush enumeration, allowing for comprehensive system evaluation (Figure 1(c)). When evaluating BIOMINER on the BIOVISTA dataset, the system achieves F1 scores of 0.22 for extracting complete bioactivity triplets, 0.45 for chemical structures (ligand coreference-SMILES), and 0.53 for bioactivity measurements (protein-ligand coreference-bioactivity value). This performance is complemented by high throughput, processing literature at approximately 14 seconds per paper on 8 V100 GPUs. Notably, BIOMINER demonstrates effectiveness in processing complex chemical representations, achieving an F1 score of 0.56 for the challenging task of Markush structure enumeration. These results validate BIOMINER's bioactivity data extraction ability and providing a quantitative baseline for future research.

We further perform three applications to showcase BIOMINER's practical utility and broad applicability. **First**, we use BIOMINER to construct a bioactivity pre-training database containing 67,953 data points extracted from 11,683 papers within two days. Models pre-trained on this BIOMINER-generated database demonstrate improved performance (3.1% and 1.9% improvement in RMSE metric) on two independent test sets (PDBbind v2016 core set¹⁹ and CSAR-HiQ²⁰) compared to models trained only on existing curated data (PDBbind v2016 refined set, 3,767 data points). **Second**, we implement a human-in-the-loop (HITL) bioactivity extraction workflow, where human experts collect data by reviewing the extraction result of BIOMINER rather than *de novo* identification and transcription, to correct errors of fully automated extraction and ensure data quality. With HITL workflow, 1,592 bioactivity data of NLRP3²¹, a high-priority target for anti-inflammatory therapies, are collected from 85 papers in 26 hours, doubling the NLRP3 data available in ChEMBL (820 data). This expanded dataset yields improved QSAR models (38.6% improvement in EF1% metric), based on which we screen molecule libraries (ChemDiv²² and Enamine²³) and identify 16 hit candidates with novel scaffolds. **Third**, besides bioactivity extraction, we utilize BIOMINER to label complex structures with reported bioactivity data, which is important in structure-based drug design for establishing

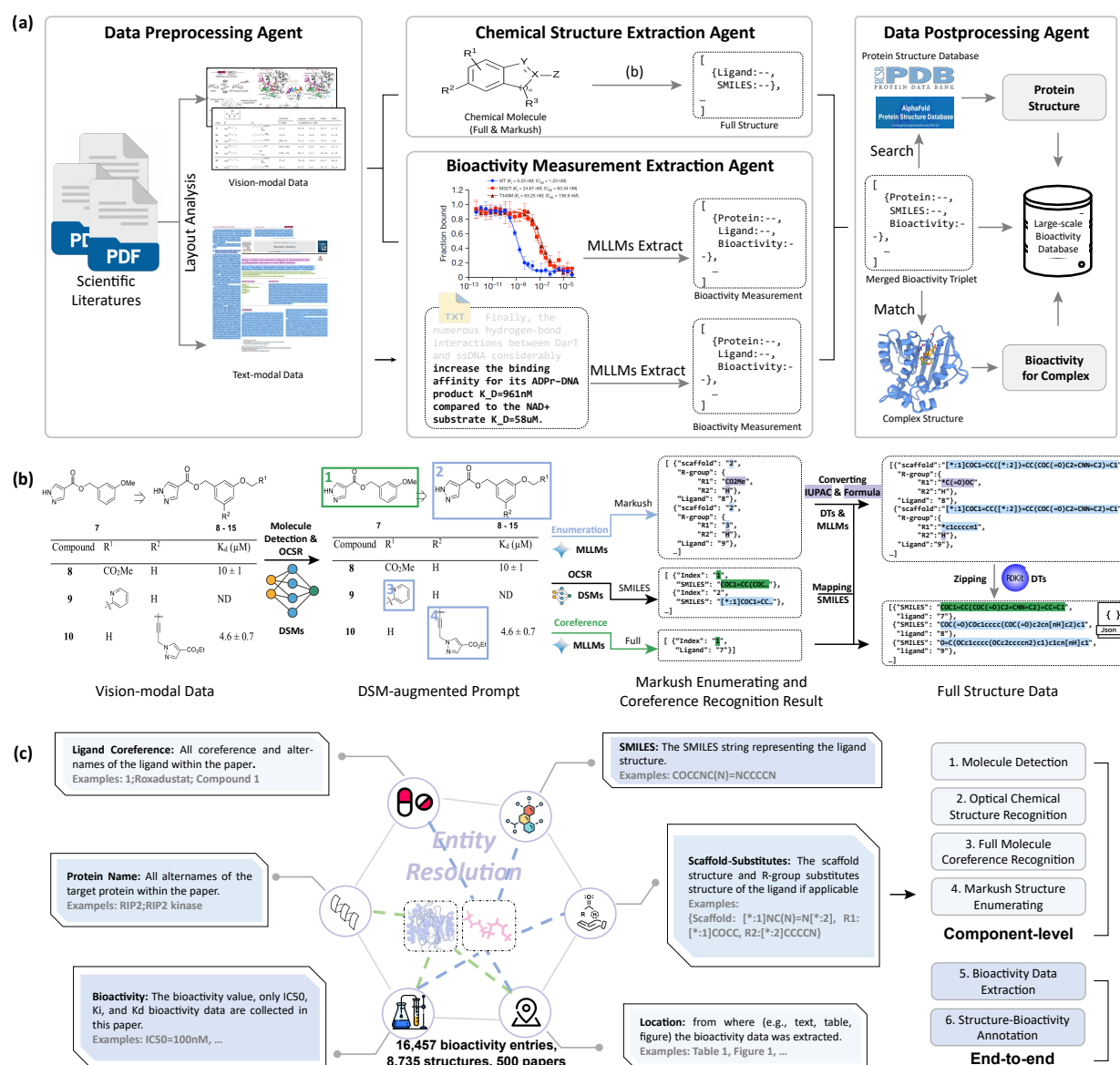


Figure 1. Overview of protein-ligand bioactivity extraction framework BIOMINER and benchmark BIOVISTA. (a) The whole protein-ligand bioactivity extraction framework BIOMINER. (b) The chemical structure extraction agent. In this agent, explicit full structures and Markush structures are both processed. For clarification, explicit full structures are plotted in green boxes and Markush scaffolds and R-group substitutes are plotted in blue boxes. (c) The new benchmark BIOVISTA, containing 16,457 bioactivity data and 8,735 structures as ground-truth labels from 500 publications. Based on these data, six evaluation tasks are designed for comprehensive evaluation, ranging from end-to-end tasks to component-level tasks.

datasets like PDBbind. Using a HITL annotation workflow similar to bioactivity extraction, we label 242 structures within the famous PoseBusters database²⁴ at an average rate of 2 minutes/structure, achieving 97% annotation accuracy—a 5-fold speedup and a 10% accuracy improvement over purely manual annotation. These experiments, taken together, demonstrate BIOMINER’s efficiency, applicability, and potential to accelerate drug discovery through both fully automated and HITL workflows. In summary, BIOMINER and BIOVISTA provide a new system and evaluation standard for automated bioactivity data extraction. These contributions offer a pathway to unlock vast amounts of previously inaccessible bioactivity data, accelerating data-driven drug discovery and establishing a foundation for future progress in automated scientific literature mining.

Results

To elucidate the performance and utility of our proposed framework, BIOMINER, we first briefly outline its architecture, with a focus on the core chemical structure extraction agent that underpins its ability to handle complex structure data (including Markush structures). Next, we introduce BIOVISTA, to the best of our knowledge, the largest benchmark designed to rigorously evaluate bioactivity extraction performance. We provide a detailed analysis of BIOMINER’s extraction performance on diverse tasks of BIOVISTA. Finally, we demonstrate BIOMINER’s practical value through three real-world applications, highlighting its versatility and impact in benefiting bioactivity data extraction and drug design.

Framework BIOMINER

The BIOMINER system, depicted in Figure 1(a), is a multi-modal system engineered for the automated extraction of protein-ligand bioactivity data from scientific literature.

Overview of BIOMINER

BIOMINER integrates four specialized agents—data preprocessing, chemical structure extraction, bioactivity measurement extraction, and data postprocessing—working collaboratively to address the complexity of this task. The key design principle of BIOMINER is tuning-free. Instead of training a novel end-to-end bioactivity extraction model, BIOMINER decomposes the complex extraction problem into four fundamental sub-tasks and leverages a synergistic combination of general multi-modal large language models (MLLMs), domain-specific models (DSMs), and domain tools (DTs) to solve them. This design eliminates the need for extensive collection of labeled data and training of task-specific models, which are common challenges in the development of specialized tools.

The extraction framework begins with the data preprocessing agent, which utilizes the DSM toolkit MinerU²⁵ to parse PDF documents. Through layout analysis and reading order determination, this agent parses both textual and visual content with high fidelity. Subsequently, the chemical structure extraction and bioactivity measurement extraction agents operate in parallel on the parsed data. The bioactivity measurement extraction agent employs the MLLM Gemini-2.0-flash²⁶, guided by carefully crafted prompts, to identify quantitative bioactivity measurements (protein-ligand coreference-bioactivity value) from text, tables, and figures. Meanwhile, the chemical structure extraction agent, detailed below, employs a MLLM-DSM-DT collaborative strategy to resolve complex chemical structures (ligand coreference-SMILES), including Markush structures. Finally, the postprocessing agent integrates the extracted bioactivity measurements from different modalities with their corresponding ligand SMILES strings to obtain the full bioactivity triplets (protein-SMILES-bioactivity value). This agent can optionally enrich the extracted data by querying external databases (e.g., UniProtKB²⁷, AlphaFoldDB²⁸, PDB²⁹) for protein structural information or annotating reported PDB complex structures with the corresponding extracted bioactivity data, thereby creating highly actionable datasets. This tuning-free pipeline—enabled by the strategic orchestration of MLLMs’ reasoning capabilities, DSMs’ domain knowledge, and DTs’ precise rules—delivers robust performance without the need for new model training, distinguishing BIOMINER as a scalable and adaptable solution in a dynamic research field.

Chemical Structure Extraction

The chemical structure extraction agent, illustrated in Figure 1(b), is a cornerstone of BIOMINER’s ability to handle the complex chemical structures of bioactivity data. It operates in three stages to extract and resolve ligand structures, including Markush structures, which are prevalent in medicinal chemistry literature and represent a significant challenge. First, DSMs (MolMiner³⁰, supplemented by YOLO³¹ for tables) detect 2D molecular structure depictions within the document. MolParser³² then performs OCSR to generate SMILES strings for these depictions. Second, using images augmented with DSM-detected structure bounding boxes and indices (Figure 1(b)), MLLM Gemini performs coreference recognition and Markush enumeration. For explicit full structure, Gemini directly recognizes its coreference, linking chemical depictions to their textual references. Critically, for Markush structures, Gemini identifies Markush scaffold structures and enumerates associated R-group substituents. Third, BIOMINER uses RDKit³³ to systematically zip the recognized Markush scaffold SMILES with the enumerated R-group substituent SMILES, generating the final list of specific, full chemical structures represented by the Markush definition. Notably, R-group substituents are often described textually (e.g., IUPAC names, abbreviations, chemical formulas) rather than visually. Before the RDKit zipping step, these 1D R-group substituents are converted into SMILES, employing OPSIN³⁴ for

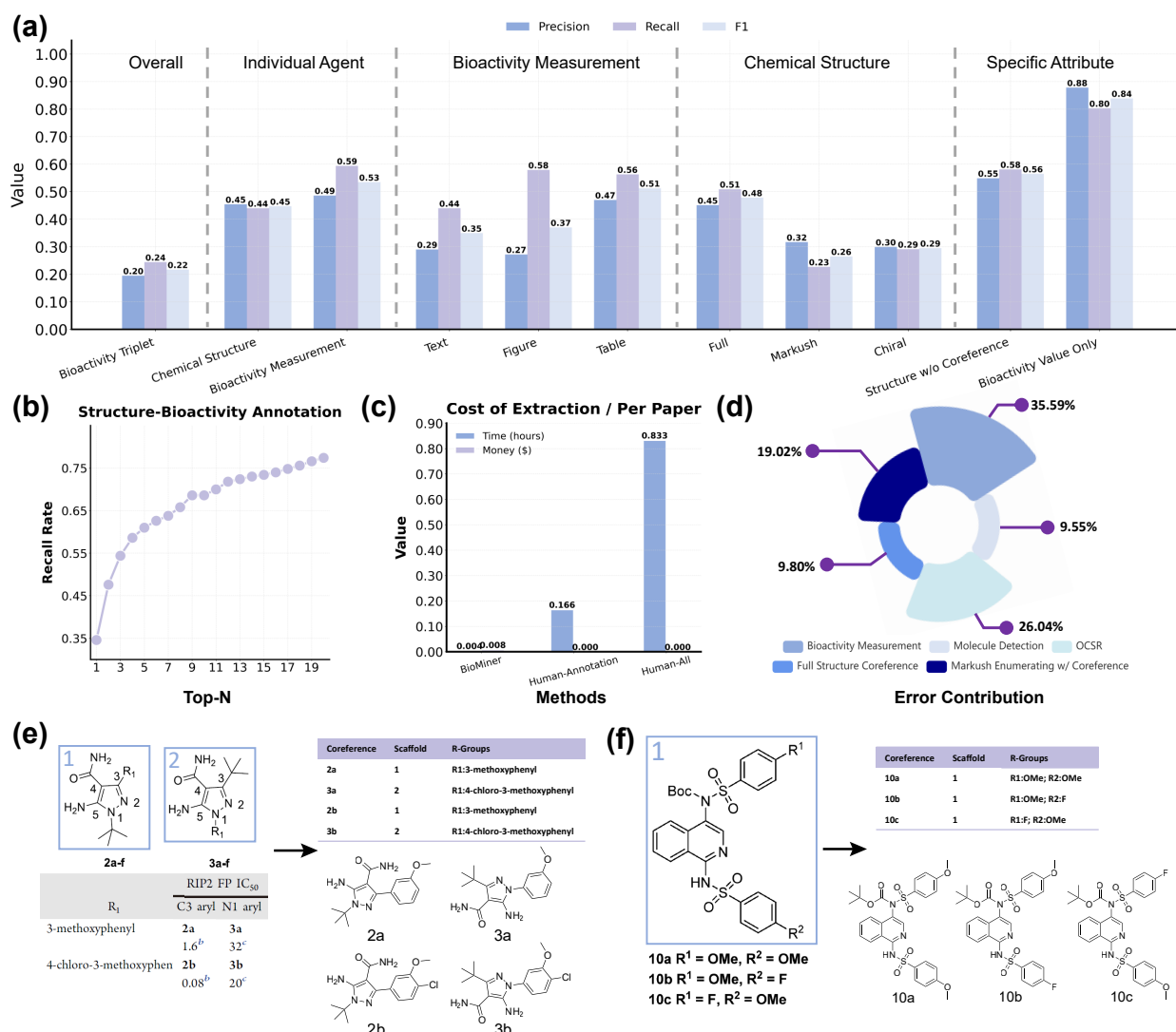


Figure 2. Benchmarking extraction performance of BIOMINER on BIOVISTA. (a) Performance of bioactivity triplet and individual attributes extraction. (b) Performance of structure-bioactivity annotation. Given a PDB structure and corresponding paper, BIOMINER first extracts all bioactivity from the paper and then ranks these data based on ligand similarity between extracted data and PDB structure. Top-N recall rate represents the ratio of target bioactivity data in Top-N extracted data. (c) Comparison on consuming time and money between BIOMINER and human. (d) Detailed error source analysis of bioactivity triplet extraction. (e,f) Two examples of Markush structures successfully processed by BIOMINER.

IUPAC names and a Gemini-assisted, manually curated mapping table for abbreviations and chemical formulas. This hybrid MLLM-DSM-DT approach, which obviates the need for task-specific model training, enables the successful resolution of both explicit full and Markush structures, representing a key innovation of BIOMINER. Implementation details of this agent are provided in the Method and Supporting Information section.

Benchmark BioVISTA and Performance Evaluation

To enable rigorous evaluation and facilitate the development of future extraction methods, we introduce BIOVISTA, a comprehensive and challenging benchmark for protein-ligand bioactivity data extraction. To our knowledge, this is the largest benchmark dedicated to the bioactivity extraction task. In this subsection, we first present the construction process of BIOVISTA, and then detail BIOMINER's performance.

Table 1. Performance of BIOMINER with different MLLMs on BIOVISTA. This table focuses on presenting MLLMs-related results and comparing the performance of different MLLMs. In BIOMINER, we choose Gemini-2.0-flash as our employed MLLM based on this table results. The best results are bolded, and the second best results are underlined.

Tasks	Metric	Gemini-2.0-flash	Gemini-2.0-pro	GPT-4o	GPT-4o-mini	Claude-3.7-sonnet	Claude-3.5-sonnet	Grok-3	Qwen-2.5
<i>Component-level evaluation for chemical structure extraction</i>									
Full Structure Coreference	Recall	0.786	<u>0.763</u>	0.749	0.552	0.749	0.755	0.529	0.748
	Precision	0.783	<u>0.760</u>	0.745	0.550	0.757	0.754	0.542	0.745
	F1	0.784	<u>0.761</u>	0.747	0.551	0.753	0.754	0.535	0.746
Markush Enumerating w/ Coreference	Recall	0.579	0.386	0.430	0.039	0.465	<u>0.477</u>	0.190	0.209
	Precision	0.545	0.377	0.426	0.041	0.475	<u>0.472</u>	0.190	0.212
	F1	0.561	0.381	0.428	0.040	0.470	<u>0.474</u>	0.190	0.210
Markush Enumerating w/o Coreference	Recall	0.628	0.425	0.500	0.050	0.503	<u>0.516</u>	0.241	0.252
	Precision	0.594	0.448	0.495	0.058	0.514	<u>0.511</u>	0.257	0.281
	F1	0.611	0.436	0.497	0.054	0.508	<u>0.513</u>	0.249	0.266
<i>Different attributes evaluation for end-to-end extraction and annotation</i>									
Structure w/ Coreference	Recall	0.440	0.374	0.406	0.277	<u>0.436</u>	0.434	0.275	0.373
	Precision	0.454	0.395	0.419	0.305	0.462	<u>0.459</u>	0.326	0.403
	F1	<u>0.447</u>	0.384	0.412	0.290	0.449	<u>0.446</u>	0.298	0.387
Bioactivity Measurement	Recall	0.594	<u>0.584</u>	0.418	0.250	0.480	0.472	0.386	0.552
	Precision	0.486	<u>0.465</u>	0.364	0.219	0.396	0.410	0.285	0.431
	F1	0.535	<u>0.518</u>	0.389	0.233	0.434	0.439	0.328	0.484
Bioactivity Triplet	Recall	0.244	<u>0.220</u>	0.171	0.088	0.208	0.208	0.109	0.199
	Precision	0.195	0.172	0.149	0.077	0.178	<u>0.185</u>	0.081	0.149
	F1	0.217	0.193	0.159	0.082	0.192	<u>0.196</u>	0.093	0.170
Bioactivity-Structure Annotation	Top-1	<u>0.346</u>	0.288	0.318	0.258	0.330	0.348	0.264	0.322
	Top-3	0.544	0.482	0.530	0.454	0.518	<u>0.538</u>	0.434	0.494
	Top-10	0.686	0.634	<u>0.688</u>	0.612	0.670	0.706	0.614	0.680
<i>Price comparison for different MLLMs</i>									
1M Tokens Cost \$	Input	0.100	1.250	2.500	<u>0.150</u>	3.300	3.300	0.830	-
	Output	0.400	5.000	10.000	<u>0.600</u>	16.500	16.500	4.160	-

Construction of BIOVISTA

BIOVISTA is derived from 500 recent publications referenced in PDBbind v2020¹², ensuring its relevance to real-world scenarios. Unlike PDBbind data curation, BIOVISTA includes all bioactivity data reported within these publications, not just the reported bioactivity to the PDB structure. Domain experts manually curate all bioactivity data points, annotating attributes beyond the core triplet (protein, ligand SMILES, bioactivity value) to enable fine-grained analysis (Figure 1(c)). These attributes (e.g., location, ligand coreference, scaffold-substitutes) include information such as the source modality to analyze the ability to process different modal data. Notably, all alternative names (alternames) for proteins and ligands are carefully collected, ensuring completeness and preventing ambiguity or overlap.

To ensure high data quality and robustness, we implement a closed-loop bootstrapping procedure. Initially, BIOMINER is used to perform automatic extractions on the curated publications, facilitating the identification and correction of potential inconsistencies or errors in the initial annotations (e.g., missing alternames, incorrect transcription). Feedback from these automatic extraction results are then incorporated into the refinement of BIOMINER's design, improving prompt engineering strategies and optimizing individual agent performance. Subsequently, the enhanced version of BIOMINER is re-applied to comprehensively validate and confirm the accuracy of the final BIOVISTA dataset. The finalized BIOVISTA dataset comprises 16,457 bioactivity data points source from text (15.8%), figures (11.6%), and tables (72.5%) (Figure S1), along with 8,735 chemical structures (of which 48.7% are derived from Markush structures).

BIOVISTA defines two end-to-end tasks and four component-level tasks to thoroughly assess the extraction performance of BIOMINER. The two end-to-end tasks evaluate overall extraction capability, specifically: (1) extracting all bioactivity data reported in a publication, and (2) annotating PDB structures with bioactivity information presented in associated papers. These tasks directly reflect the practical utility of bioactivity extraction methods. Additionally, four specialized component-level tasks—molecule detection, optical chemical structure recognition (OCSR), full-structure coreference resolution, and Markush structure enumeration—provide deeper insights into the performance of critical chemical structure extraction processes, facilitating method development and optimization. Detailed descriptions of these evaluation tasks can be found in Supporting Information Note 1. Altogether, this suite provides a robust foundation for evaluating and advancing bioactivity extraction systems. Comprehensive evaluation across the six tasks defined within the BIOVISTA benchmark rigorously assesses BIOMINER's capabilities and establishes a performance baseline for further works. Performance metrics, error analysis, cost-efficiency analysis, and case study are all included in the following part.

End-to-end Evaluation

End-to-end bioactivity extraction performance is shown in Figure 2(a), where BIOMINER achieves Precision=0.20, Recall=0.24, and F1=0.22 for the complex bioactivity triplet. In addition to the evaluation of whole bioactivity triplet, which integrates multi-modal attributes, analysis of individual attributes reveals stronger performance. Chemical structure extraction (ligand coreference-SMILES) achieves F1=0.45, with performance varying by complexity (Explicit Full F1=0.48 vs. Markush F1=0.26). Bioactivity measurement extraction (protein-ligand coreference-bioactivity value) achieved F1=0.53, with table-based extraction being the most effective (Table F1=0.51 vs. Figure F1=0.37, Text F1=0.35). Extraction of isolated attributes is even more proficient, with solely bioactivity value F1=0.84 and SMILES F1=0.56. These scores underscore the significant challenge of bioactivity extraction and indicate component strengths of BIOMINER despite integration challenges.

Performance on the other end-to-end task structure-bioactivity annotation is presented in Figure 2(b). Given a complex structure and the associated paper, BIOMINER extracts all bioactivity data from the paper, and ranks them based on ligand similarity between extracted data and given structure. When taking Top-10 bioactivity data into consideration, BIOMINER achieves a recall rate of 0.69 for the reported bioactivity. This indicates that while the Top-ranked match may not always be correct, the reported value of the given structure is often among the top candidates, supporting efficient structure-bioactivity annotation or human validation.

For end-to-end bioactivity extraction and structure-bioactivity annotation, BIOMINER offers transformative efficiency gains as indicated in Figure 2(c). It processes literature at an average rate of 0.004 hours/paper (on 8 V100 GPUs, about 14s/paper) with a very cheap cost (\$0.008/paper), orders of magnitude faster than human structure annotation and bioactivity extraction (0.166-0.833 hours/paper). These results demonstrate BIOMINER's potential for cost-effective, large-scale data mining.

Component-level Evaluation

Chemical structure extraction, which is highly related to final extraction performance and the core section of BIOMINER, is particularly evaluated on four specially designed component-level tasks. As introduced above, BIOMINER resolves ligand structures in three stages. These four component-level tasks respectively evaluate the first stage molecule detection (Table S2), OCSR (Table S3), and the second stage full structure coreference recognition, Markush enumeration (Table 1).

For first stage molecule detection, BIOMINER employs MolMiner supplemented by YOLO for table molecules, achieving mAP=0.857, AP₇₅=0.892, and AP-Small=0.185 (Table S2). Compared with directly using YOLO or MolMiner, BIOMINER's strategy achieves better performance for detecting small molecules (best AP-Small), usually R-group within tables, and meeting high Intersection over Union (IoU) requirement (best AP₇₅). As for OCSR, compared to other DSMs, the employed MolParser exhibits significantly better ability to recognize Markush structure (Accuracy=0.733 for Markush structures), and thus leading to best OCSR performance (Accuracy=0.703 for all structures) (Table S3). When it comes to second stage tasks, BIOMINER achieves an F1 score of 0.784 and 0.561 on full structure coreference recognition and Markush enumeration (with coreference) using Gemini-2.0-flash (Table 1), respectively. If we only consider the structure enumeration result and ignore coreference, F1 score of Markush enumeration rises to 0.611. This result indicates the ability of BIOMINER to process the challenging Markush structures. Different MLLMs, including Gemini, GPT³⁵, Claude³⁶, Qwen³⁷, and Grok³⁸, have been tested on these two tasks, and Gemini-2.0-flash achieves the best performance in both price and accuracy. These component-level evaluations provide a detailed analysis of extraction results and effectively benefit the design of chemical structure extraction agent.

Error Analysis and Cases Study

Based on the component-level evaluation results, we plot the error contribution analysis for the end-to-end triplet extraction in Figure 2(d). Errors in bioactivity measurement extraction contribute the largest share (35.59%), followed by OCSR inaccuracies (26.04%), and the Markush structure enumeration (19.02%). Molecule detection (9.55%) and explicit full structures coreference recognition (9.80%) contribute less to the overall error. This breakdown provides critical insights for directing future research toward optimizing the system components.

Qualitative evidence of BIOMINER's advanced capabilities is shown in Figure 2(e, f), which displays examples of complex Markush structures successfully processed by the system. This includes correct scaffold identification, R-group enumeration, 1D R-group processing (IUPAC and chemical formula), and accurate generation of the final enumerated SMILES strings, indicating its ability to process challenging Markush structures.

In summary, the benchmarking results on BIOVISTA validate BIOMINER's capacity for automated, multi-modal bioactivity data extraction. While quantifying the performance across various dimensions and pinpointing key challenges for future work, the results demonstrate certain bioactivity extraction ability and substantial efficiency advantages over manual curation. This positions BIOMINER as a valuable tool for accelerating data acquisition and unlocking previously inaccessible data in the vast body of drug discovery literature.

Automated Collection of Large-scale Bioactivity Data for Model Training

A key objective in developing BIOMINER is to overcome the limitations of existing, manually curated bioactivity databases, and to unlock the wealth of previously inaccessible data hidden within scientific literature. To demonstrate BIOMINER's capacity

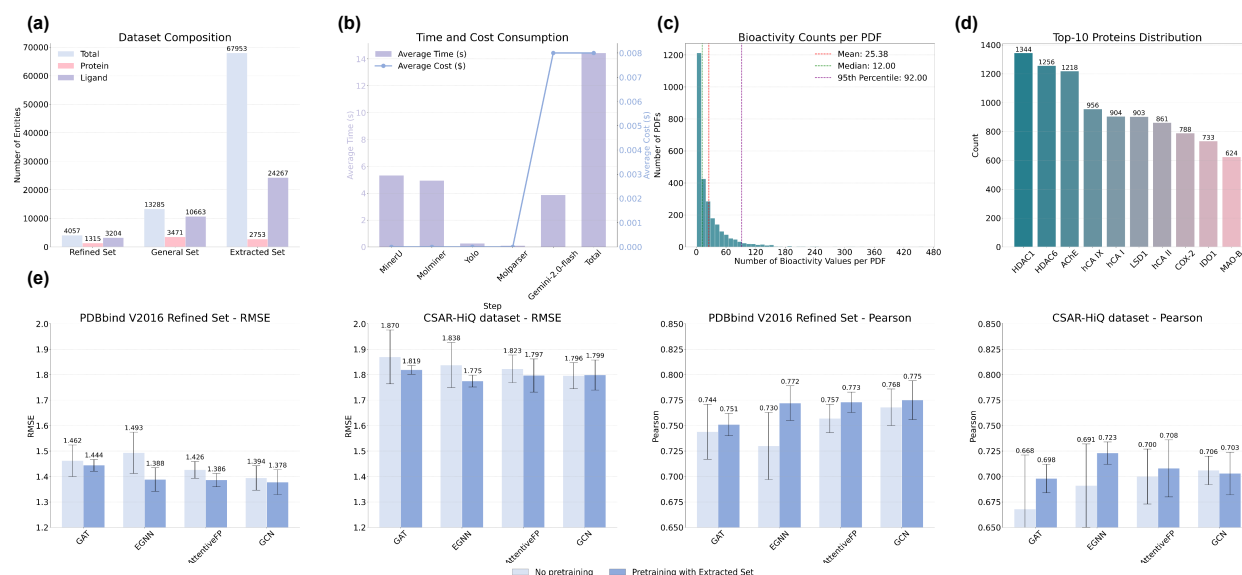


Figure 3. Using BIOMINER collecting large-scale bioactivity data for deep learning model training. (a) Statistical comparison between manually curated PDBbind v2016 dataset and our extracted dataset; (b) Consuming time and money for BIOMINER; (c) Number of extracted bioactivity data in each paper; (d) Protein names distribution within extracted bioactivity data. (e) Performance comparison of models with and without pre-training on BIOMINER extracted set.

for large-scale data acquisition, we apply it to construct a bioactivity database. Deep learning-based bioactivity prediction models are further pre-trained on this extracted database to demonstrate its utility.

To construct a large-scale dataset, we target the European Journal of Medicinal Chemistry (EJMC), a high-impact journal known for its density of protein-ligand bioactivity data, making it an ideal source. 11,683 articles published in EJMC since 2010 are collected, excluding earlier articles due to potential low resolution. Employing BIOMINER, we process the 11,683 papers within just two days (about 14s/paper on 8 V100 GPUs), a speed that would be practically impossible with manual curation (Figure 3(b)). From these paper, 195,046 bioactivity triplets are extracted. To support the training of bioactivity prediction models, we further enrich these data with protein structure information. As discussed above, using the extracted protein name, BIOMINER searches external structure databases (including AlphaFoldDB and PDB) for protein structure information. 67,953 data points are successfully enriched with protein structure information, providing a large-scale extracted database significantly expands the available data compared to PDBbind's refined and general sets (Figure 3(a)). After excluding paper contains no bioactivity data, analysis of the extracted dataset reveals a mean of 25.38 bioactivity values per paper, with a median of 12 and a 95th percentile of 92, highlighting the density of information contained within individual publications (Figure 3(c)). Further analysis of the extracted dataset shows the Top-10 protein distributions, indicating the most researched proteins and their potential application in bioactivity prediction (Figure 3(d)).

Critically, we evaluate the impact of this automatically extracted database on the performance of downstream deep learning models. We pre-train several graph neural network (GNN) architectures (GAT³⁹, EGNN⁴⁰, AttentiveFP⁴¹, and GCN⁴²) on the database constructed by BIOMINER and compared their performance against models trained solely on the PDBbind v2016 refined set (3,767 data points), across two independent test sets: PDBbind v2016 core set and CSAR-HiQ set. As demonstrated in Figure 3(e), models pre-trained on the BIOMINER-derived database consistently achieved performance improvements, as measured by both Root Mean Squared Error (RMSE) and Pearson correlation coefficient. Average improvements of 3.1% and 1.9% in the RMSE metric are achieved in the PDBbind core set and CSAR-HiQ set, respectively. These results underscore the value of automatically extracted bioactivity data for enhancing the accuracy and predictive power of computational models in drug discovery. BIOMINER can serve as an efficient and cost-effective method for mining scientific literature, which is crucial for data-driven AI-powered drug design. Details about model training can be found in Supporting Information.

Human-in-the-loop Curation of NLRP3 Bioactivity Data and Enhanced Inhibitor Screening

While fully automated extraction using BIOMINER demonstrates high throughput, potential data errors will be introduced. In practical scenario, correcting potential errors and ensuring high data quality is crucial for sensitive downstream applications like quantitative structure-activity relationship (QSAR) model building and drug screening. To address this, in this subsection,

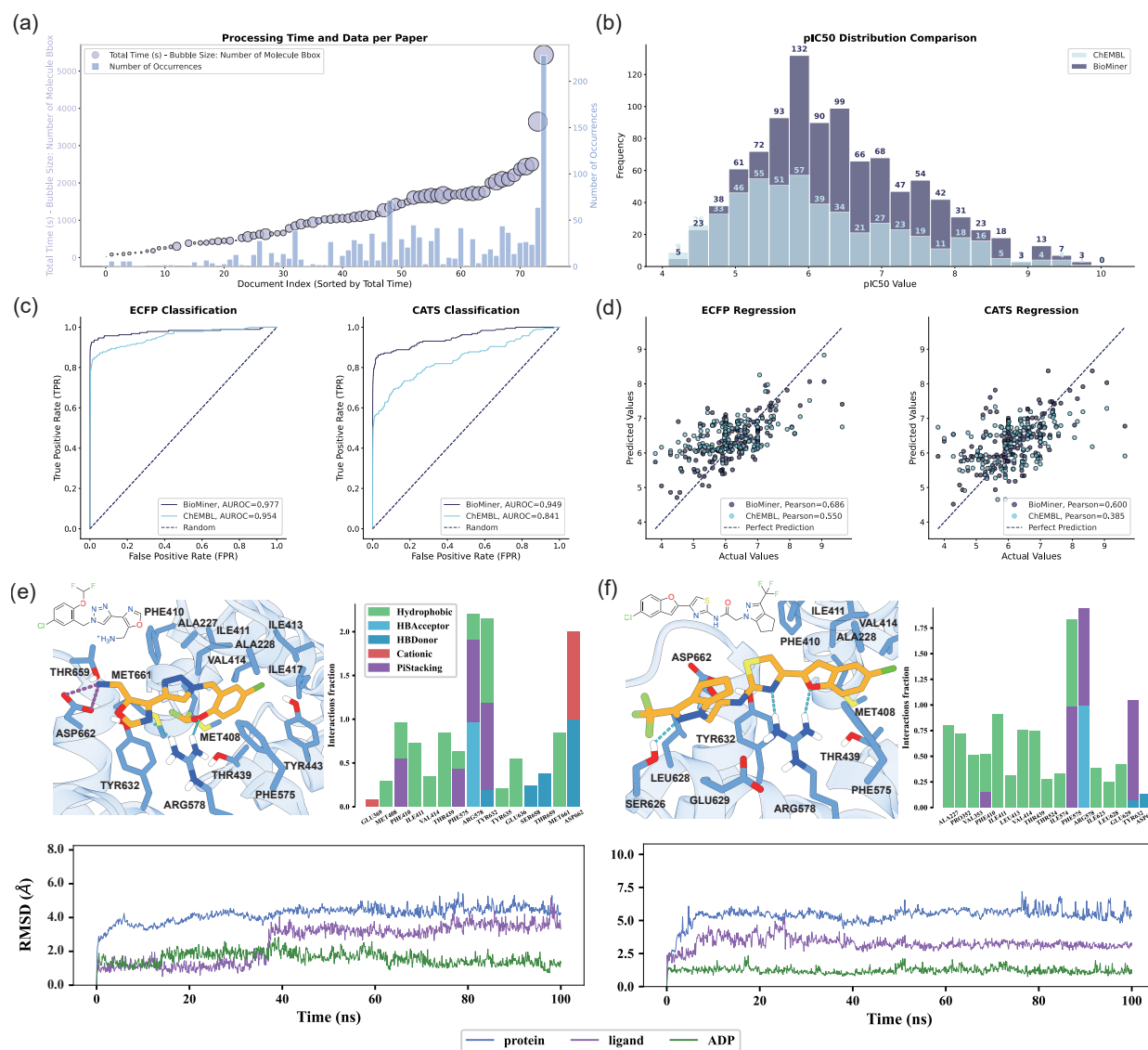


Figure 4. Using BIOMINER for human-in-the-loop NLRP3 bioactivity data collection from 85 papers and inhibitor screening. (a) Consuming time (about 18.4 minutes in average) and number of collected data for each paper. The final consuming time is highly related with the number of bioactivity data and chemical structures; (b) Comparison of pIC50 distribution between BIOMINER collected bioactivity data and ChEMBL data; Classification (c) and regression (d) performance comparison between QSAR models trained on BIOMINER data and ChEMBL data; The Glide-XP docked binding pose within the NP3-253-binding pocket (PDBID: 9GU4), stacked bar plot of the interaction fraction during MD simulation and RMSD curve of protein, ligand and ADP for Z6739936901 (e) and Z5232931194 (f), respectively.

we introduce a human-in-the-loop (HITL) workflow to enable high-quality and efficient data collection aided by BIOMINER and validate its effectiveness in a practical task.

Bioactivity Data Collection HITL Workflow

In the HITL workflow, human experts extract bioactivity data by reviewing and validating the chemical structure and bioactivity measurement extraction results of BIOMINER, which consists the final bioactivity triplet. For chemical structures, as shown in Figure S2, S3, and S4, human expert verifies the first stage molecule detection, OCSR, and the second stage full structure coreference recognition, Markush enumeration, sequentially. For bioactivity measurement, the expert directly confirms the correctness of the extracted protein-ligand coreference-bioactivity value tuples, the output of the bioactivity measurement extraction agent. Critically, human intervention is targeted only at these verification steps. Other components within BIOMINER, such as 1D R-group processing and Markush scaffold-R-group zipping, are rule-based steps that do not require manual review. Overall, this HITL workflow omits the *de novo* identification and transcription of chemical structure and bioactivity, and experts primarily focus on verifying BIOMINER's outputs, significantly accelerating curation compared to fully manual extraction.

NLRP3 Inflammasome Bioactivity Data Collection

To showcase this workflow's practical utility, we apply it to curate NLRP3²¹ inflammasome bioactivity data. NLRP3 is a high-priority target for anti-inflammatory therapies, yet publicly available bioactivity data, such as in ChEMBL, remains relatively sparse. Our objective was to expand the size of high-quality NLRP3 bioactivity dataset and utilize this enriched data to develop more accurate models for inhibitor discovery.

We identify 85 relevant scientific publications reporting NLRP3 bioactivity data through literature searches and then collect 1,592 data from these publications with this HITL workflow. As shown in Figure 4(a), the time required per paper varied, correlating with the number of chemical structures and bioactivity data points present. On average, it takes approximately 18.4 minutes for each paper and consumes 26 hours for all 85 papers. This effort effectively doubles the amount of NLRP3 bioactivity data previously available in ChEMBL (Figure 4(b)), providing a substantially larger and chemically diverse dataset. The expert oversight inherent in the HITL process also allows for meticulous handling of ambiguities and edge cases, further ensuring data quality.

QSAR Models Training

To assess the impact of this data expansion, we train quantitative structure-activity relationship (QSAR) models for predicting NLRP3 inhibition using both the original ChEMBL NLRP3 dataset and our expanded, BIOMINER-curated NLRP3 dataset, respectively. As shown in Table S5 and Table S6, various QSAR models with different algorithms (e.g., Random Forest⁴³, SVM⁴⁴, etc.), different tasks (regression, classification), and different molecular representations (ECFP⁴⁵, CATS⁴⁶) are evaluated comprehensively. Models trained on the BIOMINER-curated dataset consistently outperformed those trained on the standard ChEMBL data across all settings evaluated. On average, considering the EF1% metric across 28 distinct QSAR model configurations, the BIOMINER-curated dataset yielded a 38.6% performance improvement. Figure 4(c) and (d) visualize the performance of the top-performing models under various settings. Notably, for classification using ECFP fingerprints, the AUROC improved from 0.954 (ChEMBL) to 0.977 (BIOMINER-curated). Similarly, for CATS-based regression, the Pearson correlation coefficient increased substantially from 0.385 to 0.600. This marked enhancement in predictive performance significantly increases the reliability of bioactivity prediction for novel chemical entities, thereby improving prospects for effective inhibitor screening.

NLRP3 Inhibitor Screening

Leveraging the superior QSAR models trained on the BIOMINER-curated data, we initiate a virtual screening against chemical libraries (i.e., ChemDiv and Enamine). Through our established rational screening pipeline (Figure S8 and see Supplementary Information for details), sixteen compounds are manually selected as potential hit candidates, exhibiting both high structural diversity and novelty (Table S7). Further MM/PBSA binding free energy calculation reveals six compounds exhibit comparable or better binding free energies than both MCC950⁴⁷ and NP3-562⁴⁸, suggesting their potential as highly potent NLRP3 inhibitors (Table S7). Docking simulation demonstrates two promising candidates, Z6739936901 and Z5232931194, showing considerable binding mode between the NBD, HD1, WHD, and HD2 domains (Figure 4(e,f)). MD simulation of 100 ns is performed to investigate the binding stability of identified compounds. The RMSD of protein and ligand shows that Z6739936901 and Z5232931194 bind stably with the binding pocket in the last 60 ns. Z6739936901 maintains persistent hydrogen bonding, cation- π and π - π interactions with ARG578 and TYR632. Its solvent-exposed charged amino group forms strong salt bridges and hydrogen bonds with ASP662, anchoring the oxazole moiety rigidly. Meanwhile, the phenyl fragment remains tightly bound within a large hydrophobic subpocket. Z5232931194 frequently forms hydrogen bonds with ARG578, PHE575, and TYR632, along with cation- π and π - π interactions. The bromine-substituted benzofuran moiety remains tightly bound within the hydrophobic subpocket. Notably, the ADP bound in the substrate pocket remains stable throughout MD simulations, with low RMSD values (<1.5 Å) confirming its rigid positioning with minimal conformational changes. This

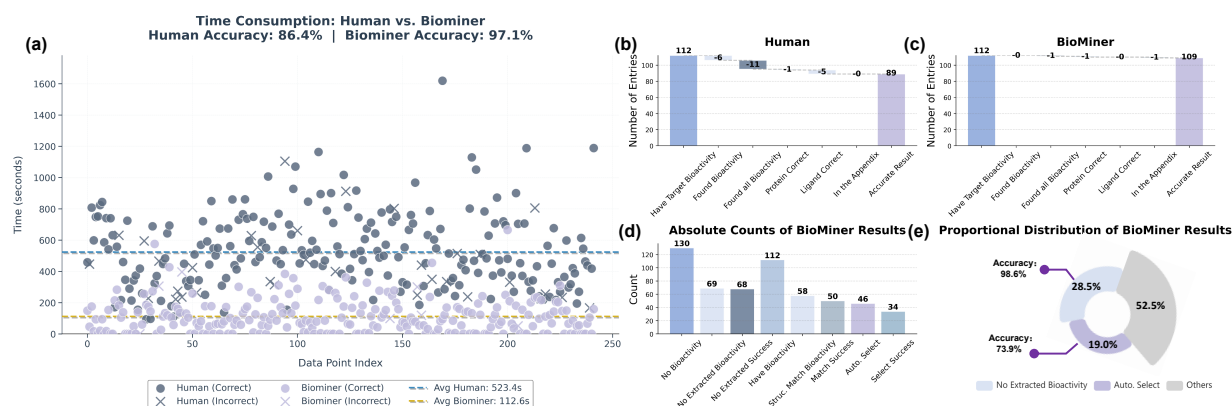


Figure 5. Annotating protein-ligand complex structures in PoseBusters with corresponding experimental bioactivity values. (a) Comparison of consuming time and annotation accuracy between human and BIOMINER; (b) Waterfall plot of error analysis of human extraction; (c) Waterfall plot of error analysis of BIOMINER extraction; (d) Number analysis of different data of BIOMINER results; (e) Proportional analysis of different data of BIOMINER results.

observation suggests that both Z6739936901 and Z5232931194 effectively stabilize the inactive conformation of the NBD, HD1, WHD, and HD2 domains. Collectively, our BIOMINER-curated data-augmented QSAR models identifies several structurally novel hit candidates exhibiting high potential as potent NLRP3 inhibitors. These candidates represent promising chemotypes for future experimental validation, potentially unlocking new avenues in NLRP3-targeted drug discovery.

In summary, this study demonstrates the effectiveness of a BIOMINER-powered HITL workflow for rapidly constructing large, high-quality, target-specific bioactivity datasets. The efficiency gain (approx. 18.4 min/paper) over manual curation, combined with the significant improvement in downstream QSAR model performance, underscores the potential of this approach to accelerate data-driven drug discovery efforts.

Enhancing Structure-Bioactivity Annotation for PoseBusters

Besides bioactivity data extraction, annotating experimentally determined protein-ligand complex structures with their reported bioactivity measurements in the literature is also important in structure-based drug design field for establishing datasets like PDBbind. In the experiments on the BIOVISTA, BIOMINER has demonstrated its capability to perform this task. The reported bioactivity measurement of a given structure is often found among the top candidates of the extracted bioactivity data (Top-10 Recall=0.686). In this subsection, we further demonstrate BIOMINER's utility in a practical annotation scenario, proposing a HITL annotation workflow and presenting a detailed analysis for fully automated data annotation beyond Top-N Recall.

Structure-Bioactivity Annotation HITL Workflow

In the annotation HITL workflow, given a PDB complex structure and associated publication, BIOMINER first extracts all potential protein-ligand-bioactivity triplets and ranks extracted data based on ligand similarity between extracted SEMILS and PDB ligand SMILES. Then, a human expert rapidly verifies the bioactivity data in the ranked list one by one. Compared with purely human annotation, such a HITL streamlines the expert's task to validation rather than *de novo* annotation.

PoseBusters Structures-Bioactivity Annotation

The famous PoseBusters benchmark²⁴, which contains 308 high-quality complex structures, is selected for HITL annotation. After careful preparation of the input, 242 PDB structures and associated publications are collected. To highlight our workflow's ability, the annotation results between a human expert using the HITL workflow and the other human expert without it are compared. As shown in Figure 5(a), this BIOMINER-assisted HITL workflow demonstrates significant efficiency gains, requiring an average of only about 2 minutes per publication for verification and annotation. The result represents an estimated 5-fold acceleration compared to purely manual curation efforts. Crucially, the accuracy of associating the correct bioactivity value with the specific complex structure improves substantially to 97.1%, compared to an estimated 86.4% accuracy baseline for manual annotation alone. An analysis of annotation errors (Figure 5(b,c)) reveals that manual annotation is prone to overlooking the correct bioactivity data (failed to find or cannot find all correct data), a category of error effectively mitigated by BIOMINER's systematic extraction and presentation of candidates. The improved accuracy of the HITL approach can thus be attributed to BIOMINER's comprehensive extraction combined with expert validation of a focused, ranked list, leading to a more robust and efficient annotation process.

Fully Automated Annotation

Based on the HITL annotation result, we further analyze the performance of the fully automated annotation beyond the Top-N Recall (Figure 5(d,e)). For 69 of 242 complexes, BIOMINER finds no bioactivity data in the publications. Among these complexes, 68 complexes are correct (consistent with manual verification confirming absence). For the remaining 173 complexes where bioactivity data are potentially present, BIOMINER successfully extracts candidate bioactivity measurements. Within this group, 58 complexes have at least one extracted bioactivity measurement associated with a ligand structure perfectly matching the PDB ligand (chemical similarity = 1.0). For these 58 matches, the data postprocessing agent (utilizing the Gemini MLLM) selects the most probable bioactivity value based on contextual consistency with the protein name and PDB structure title. This automated selection process identifies a candidate value in 46 cases. Manual validation confirms that 34 of these 46 automatically selected annotations are correct, yielding an accuracy of 73.9% for this specific subset. Combining the 69 cases correctly identified as having no data with the 46 cases where a candidate value is proposed, BIOMINER demonstrates reliable automated handling for 115 out of 242 structures (47.5%). This suggests that nearly half of the annotation task for this dataset could potentially be automated with high confidence (Accuracy=0.887), significantly reducing manual effort, while the remaining cases benefit from the efficient HITL verification.

The ability to rapidly generate and verify high-fidelity structure-activity annotations for large structural datasets like PoseBusters is highly valuable for the development of structure-based drug design algorithms. By significantly accelerating the creation and curation of these critical datasets with enhanced accuracy, BIOMINER directly facilitates progress in structure-guided drug discovery and the application of machine learning in structural biology.

Discussion

In this work, we introduced BIOMINER, a multi-modal system leveraging the synergistic combination of multi-modal large language models (LLMs), domain-specific models (DSMs), and domain tools (DTs) to automate the extraction of protein-ligand-bioactivity triplets from scientific papers. To rigorously evaluate such systems and guide future development, we established BIOVISTA, the largest bioactivity extraction benchmark featuring 16,457 bioactivity entries and 8,735 chemical structures across six distinct evaluation tasks.

On BIOVISTA, BIOMINER demonstrated considerable extraction capabilities, achieving F1 scores of 0.22, 0.45, and 0.53 for bioactivity triplets, chemical structures, and bioactivity measurements, respectively, alongside high operational efficiency (14s/paper on 8 V100 GPUs). The practical significance of BIOMINER was further underscored by three applications. Its scalability enabled the rapid generation of a large-scale bioactivity database (67,953 data points from 11,683 papers in 2 days), leading to improvements (e.g., 3.1% RMSE reduction) in downstream deep learning models pre-trained on this data (Figure 3). Moreover, human-in-the-loop (HITL) workflows showcased the powerful synergy between BIOMINER and human expertise. This approach doubled the high-quality NLRP3 bioactivity data compared to ChEMBL in just 26 hours, resulting in markedly improved QSAR models (38.6% EF1% gain) and the identification of 16 novel hit candidates (Figure 4). Similarly, HITL annotation of the PoseBusters benchmark yielded a 5-fold speedup and enhanced accuracy (97.1% vs. 86.4% manual baseline) (Figure 5), highlighting BIOMINER's value in curating crucial structural datasets.

While BIOMINER marked significant progress, the modest end-to-end F1 score (0.22 for triplets) underscored the inherent complexity of integrating information across multiple modalities, interpreting complex chemical representations like Markush structures, and accurately linking entities scattered throughout publications. Our error analysis (Figure 2(d)) pinpointed specific bottlenecks demanding further refinement: reliably extracting and associating bioactivity measurements from diverse formats (35.6% error contribution), improving the robustness of OCSR (26.0%), and enhancing the handling of Markush structure enumeration (19.0%). Additional limitations included an intrinsic dependency on the rapidly evolving capabilities of underlying MLLMs. Future work should therefore prioritize advancing cross-modal reasoning mechanisms, strengthening component integration, and potentially exploring targeted fine-tuning or domain adaptation strategies to optimize performance on critical sub-tasks while maintaining adaptability.

Looking forward, BIOMINER's modular architecture and its proven ability to handle complex chemical information suggested significant potential for generalization. The system's core capabilities in multi-modal integration, entity recognition, relationship extraction, and chemical structure processing could be readily adapted related extraction tasks in cheminformatics and bioinformatics, such as mining RNA-ligand interactions or molecule ADMET properties. Tailoring specific agents, such as the bioactivity measurement extractor or the chemical structure extractor for different molecular types or assay readouts, while leveraging the established framework for document parsing, multi-modal integration, and coreference resolution, would be key to extending BIOMINER's utility.

In conclusion, BIOMINER represented a significant advance in automated bioactivity data extraction, complemented by BIOVISTA as an essential standard for rigorous evaluation and comparison. Together, they offered a scalable pathway to unlock vast amounts of previously inaccessible bioactivity data trapped in literature, thereby accelerating data-driven drug discovery and laying a robust foundation for future progress in automated scientific information extraction.

Methods

Preliminary

For notation, we represent a literature document as D . Each protein-ligand bioactivity data point b in the document is represented as a tuple $b = (p, l, v)$ with three attributes: the protein target name p , the ligand SMILES string l , and the bioactivity value v . Denoting the set of bioactivity data points within the literature as $B = \{b_1, b_2, \dots, b_n\}$, where n is the number of bioactivity data points. The target of BIOMINER can be formally written as:

$$B = \text{BIOMINER}(D). \quad (1)$$

To achieve this, BIOMINER comprises four specialized agents: the data preprocessing agent A^{pre} , the chemical structure extraction agent A^{struc} , the bioactivity measurement extraction agent A^{measr} , and the data postprocessing agent A^{post} . In the following, we detail the functionality of each agent. *Additional information, including ablation study, MLLMs' prompt, technical tricks, and model implementations, can be found in the Supporting Information section.*

Data Preprocessing Agent

The data preprocessing agent A^{pre} initiates the pipeline by parsing the multi-modal content of document D , aiming to preserve essential information while minimizing noise for subsequent extraction tasks. To accomplish this, agent A^{pre} first employs the MinerU toolkit²⁵, which integrates a suite of models for document parsing, including layout analysis and reading order determination. This operation is formalized as:

$$D^{text}, D^{layout} = \text{MinerU}(D), \quad (2)$$

where D^{text} represents the extracted text containing all information for extracting bioactivity measurement in text modality, and D^{layout} denotes the layout analysis output, comprising categorized elements (e.g., text block, figure body, figure caption) and their corresponding regions within each page of D .

The agent further processes D to prepare visual data for downstream analysis. Pages of the document are converted to images D_{page}^{vision} . Then, based on D^{layout} , figures and tables are segmented from the document. To ensure completeness, only when captions are detected, the tables or figures are segmented out; otherwise, the entire page is treated as a single image. This process yields the visual data subset D_{seg}^{vision} , which encapsulates figures, tables, and their associated metadata. Here, D_{page}^{vision} supports molecule detection (not yet implemented in MinerU), and D_{seg}^{vision} facilitates Markush structure enumeration and full structure coreference recognition by isolating irrelevant text surrounding. The output of A^{pre} is thus a dual-modality representation of the document, defined as:

$$D^{text}, D^{vision} = A^{pre}(D), \quad (3)$$

where $D^{vision} = (D_{page}^{vision}, D_{seg}^{vision})$. This structured parsing into textual D^{text} and visual D^{vision} components provides a robust foundation for the extraction of bioactivity measurements and chemical structures, as described in the following subsections.

Chemical Structure Extraction Agent

The chemical structure extraction agent A^{struc} is tasked with detecting and recognizing chemical structures within a document, producing a set of ligand coreferences and their corresponding SMILES strings. This is formalized as:

$$(c_i, l_i) = A^{struc}(D^{vision}), i \in \{0, 1, \dots, K^{struc} - 1\}, \quad (4)$$

where c_i is the ligand coreference (e.g., textual name), l_i is the SMILES string, and K^{struc} is the number of structures.

Chemical structures in scientific literature generally appear as either explicit full structures or Markush structures¹⁸. Explicit full structure is standalone molecule requiring only coreference resolution to link them to textual mentions. Markush structures, however, represent families of related compounds, typically with variable substitutions (R-groups), variable attachment positions, or repeating substructures (Figure S6). This work focuses on Markush structures with variable substitutions, where the Markush scaffold is representable as a SMILES string, as other types (e.g., variable positions or repeats) are incompatible with SMILES notation. To tackle this challenging structure extraction task, A^{struc} employs a novel MLLM-DSM-DT hybrid strategy that splits the extraction into three stages, as shown in Figure 1(b).

Molecule Detection and OCSR. The process begins with DSMs detecting 2D molecular graphs in D^{vision} and converting them to 1D SMILES. Both explicit full structures and Markush structures (including both scaffolds and R-group substituents) are detected and processed. The distinction between full structures and Markush structures is made based on the presence of attachment points in the recognized SMILES strings. This operation is expressed as:

$$l_i^{full}, l_i^{part}, x_j^{full}, x_j^{part} = \text{DSM}(D^{vision}), i \in \{0, 1, \dots, K^{full} - 1\} \text{ and } j \in \{0, 1, \dots, K^{part} - 1\}, \quad (5)$$

where l_i^{full} , x_i^{full} , and K^{full} denote ligand SMILES strings, bounding boxes, and numbers for full structures (no attachment points), and l_j^{part} , x_j^{part} , K^{part} for Markush components (with attachment points). MolMiner³⁰ detects molecules across pages D_{page}^{vision} , supplemented by YOLO³¹ specially for segmented tables D_{seg}^{vision} , addressing MolMiner's tendency to merge closely spaced structures. MolParser³², a recently proposed model can handle different plot styles of Markush attachment points based on its novel end-to-end training paradigm (Table S2), then performs OCSR.

Markush Enumerating and Coreference Recognition. With detected structures, we first create DSM-augmented images, $D^{aug} = (D^{aug,full}, D^{aug,part})$, by plotting the bounding boxes onto the original images and adding unique numerical identifiers (artificial indices) to each box (top-left corner), as shown in Figure 1(b). Here, the creation of DSM-augmented images leverages D_{seg}^{vision} to isolate irrelevant text surrounding. These augmented images are processed separately for explicit full structures and Markush structures by MLLM Gemini-2.0-flash²⁶.

For explicit full structures, Gemini is prompted to identify the coreference corresponding to each numbered bounding box. The output is a set:

$$(c_i^{full}, \hat{c}_i^{full}) = Gemini(D^{aug,full}, T^{full}), i \in \{0, 1, \dots, K^{full} - 1\}, \quad (6)$$

where c_i^{full} is the recognized coreference for a full structure box, $\hat{c}_i^{full}$ is the artificial index, and T^{full} represents the prompt for this full structure process step. For Markush structures, Gemini is prompted to identify Markush scaffolds, determine if they represent variable substitutions, and enumerate all R-group substituents for each scaffold in. The output is:

$$(c_i^{part}, \hat{c}_i^{part,scaf}, \hat{c}_i^{part,R}) = Gemini(D^{aug,part}, T^{part}), i \in \{0, 1, \dots, \hat{K}^{full} - 1\}, \quad (7)$$

where c_i^{part} is the coreference for the full structure derived by combining Markush scaffold and R-group substitutes, $\hat{c}_i^{part,scaf}$ is the artificial index of this full structure's scaffold box, $\hat{c}_i^{part,R}$ is the set of R-group substitutes, T^{part} is the prompt for this markush structure process step, and $\hat{K}^{full}$ is the number of combination of Markush scaffolds and R-group substitutes. Each element in $\hat{c}_i^{part,R}$ is the artificial index of a 2D R-group depiction or the 1D textual representation of R-group (IUPAC name, abbreviation, or chemical formula).

Substitutes Processing and Full Structure Conversion. For explicit full structure, we can easily obtain the ligand coreference-SMILES pair (c_i^{full}, l_i^{full}) by replacing artificial index $\hat{c}_i^{full}$ in coreference recognition result $(c_i^{full}, \hat{c}_i^{full})$ as its SMILES. Then for Markush structures $(c_i^{part}, \hat{c}_i^{part,scaf}, \hat{c}_i^{part,R})$, substituents in $\hat{c}_i^{part,R}$ are converted to SMILES: IUPAC names via OPSIN³⁴, and abbreviations or formulas via a manually curated mapping table assisted by Gemini. RDKit³³ then combines scaffold and substituent SMILES to yield (c_i^{part}, l_i^{part}) . Both explicit structure and Markush structure sets are merged into the final output (c_i, l_i) .

Bioactivity Measurement Extraction Agent

The agent A^{measr} is design to extract bioactivity measurement data (p, c, v) from text and vision data using MLLMs:

$$(p_i, c_i, v_i) = A^{measr}(D^{text}, D_{page}^{vision}), i \in \{0, 1, \dots, K^{measr} - 1\}, \quad (8)$$

where K^{measr} is the number of bioactivity measurement. Gemini is utilized to processes text and visual data separately:

$$(p_i^{text}, c_i^{text}, v_i^{text}) = Gemini(D^{text}, T^{text}), i \in \{0, 1, \dots, K^{measr,text} - 1\}, \quad (9)$$

$$(p_i^{vision}, c_i^{vision}, v_i^{vision}) = Gemini(D_{page}^{vision}, T^{vision}), i \in \{0, 1, \dots, K^{measr,vision} - 1\}, \quad (10)$$

where T^{text} and T^{vision} are prompts designed for text and vision modality respectively. As both outputs share a consistent format, they are merged into (p_i, c_i, v_i) .

Data Postprocessing Agent

The data postprocessing agent A^{post} integrates bioactivity measurements (p_i, c_i, v_i) and full chemical structures (c_i, l_i) via ligand coreference, yielding triplets (p_i, l_i, v_i) . To enhance utility for downstream tasks (e.g., bioactivity prediction), beyond the literature itself, A^{post} enriches these with protein and complex structure data from external databases. To this end, beyond this triplet, A^{post} further enriches the bioactivity data with protein structure information or even protein-ligand complex structure, providing actionable information for downstream analysis.

Protein Structure Retrieval. A^{post} enriches the protein structure information by searching protein structure database based on extracted protein name p_i . It leverages Gemini's knowledge to filter invalid entries, then queries PDB¹² and UniProtKB²⁷ APIs to retrieve experimental co-crystal or AlphaFold2-predicted structures⁴⁹.

Structure-Bioactivity Annotation. If the document has a corresponding PDB complex structure, A^{post} attempts to identify the complex structure's bioactivity data point within the extracted set. It compares the SMILES string of the ligand in the PDB entry to the SMILES strings of all extracted ligands. Exact matches (ligand similarity = 1.0) trigger protein name comparison using Gemini, which selects out the corresponding bioactivity data based on PDB titles and extracted protein names p_i . The final output of A^{post} is an enriched set of bioactivity data triplets, including protein structure information and, when available, links to PDB complex structures.

Overall, BIOMINER synergistically combines MLLMs' reasoning, DSMs' domain expertise, and DTs' precise rules to deliver a scalable, zero-shot solution for bioactivity data extraction, requiring no task-specific training.

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Author contributions statement

Conceptualization: J.X.Y., X.K.L., J.T.Z., Z.X.Z., and Q.L.; Data Curation: J.X.Y., K.Z., Y.H.Y., and B.Y.Z.; Formal Analysis: J.X.Y., Y.H.Y., K.Z., Z.X.Z., J.T.Z., and X.K.L.; Investigation: J.X.Y., and J.T.Z.; Methodology: J.X.Y., X.K.L., Z.X.Z., and J.T.Z.; Software: J.X.Y., Y.H.Y., J.T.Z., and J.C.X.; Validation: J.X.Y., J.T.Z., K.Z., Z.X.Z., and Q.L.; Visualization: J.X.Y., and J.T.Z.; Writing – Original Draft: J.X.Y., and J.T.Z.; Writing – Review & Editing: J.X.Y., J.T.Z., Z.X.Z., K.Y.G., and Q.L.; Supervision: Z.X.Z., K.Z., and Q.L.; Project administration: K.Z., and Q.L.; Resources: K.Z., K.Y.G., and Q.L.; Funding Acquisition: Q.L.; All authors reviewed the manuscript.

Data availability

The proposed benchmark BIOVISTA is now available in <https://github.com/jiaxianyan/BioMiner>. The PDBbind dataset is available in <http://www.pdbbind.org.cn/>. The ChEMBL dataset is available in <https://www.ebi.ac.uk/chembl/>. The molecule library ChemDiv is available in <https://www.chemdiv.com/>. The molecule library Enamine is available in <https://enamine.net/>. The PoseBuster dataset is available in <https://github.com/maabuu/posebusters>.

Code availability

All codes of proposed BIOMINER are released in <https://github.com/jiaxianyan/BioMiner>.

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